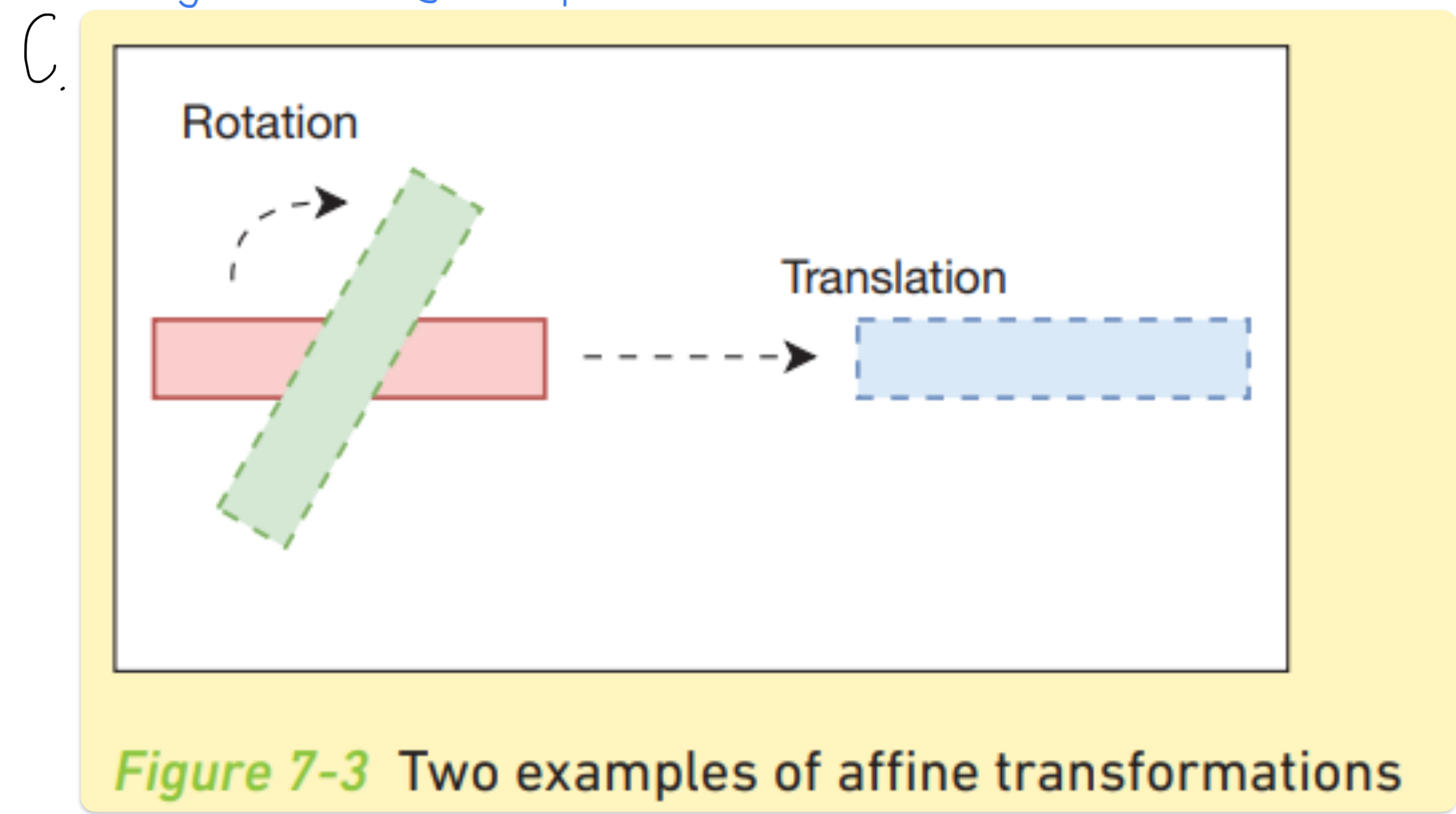


1. Translation Invariance:

- A. this means that even if an object in an image is shift horizontally or vertically, the network can still correctly identify it.
- B. a key property of convolutional layers is translation invariance, and it caused by weight sharing and sparse connections.

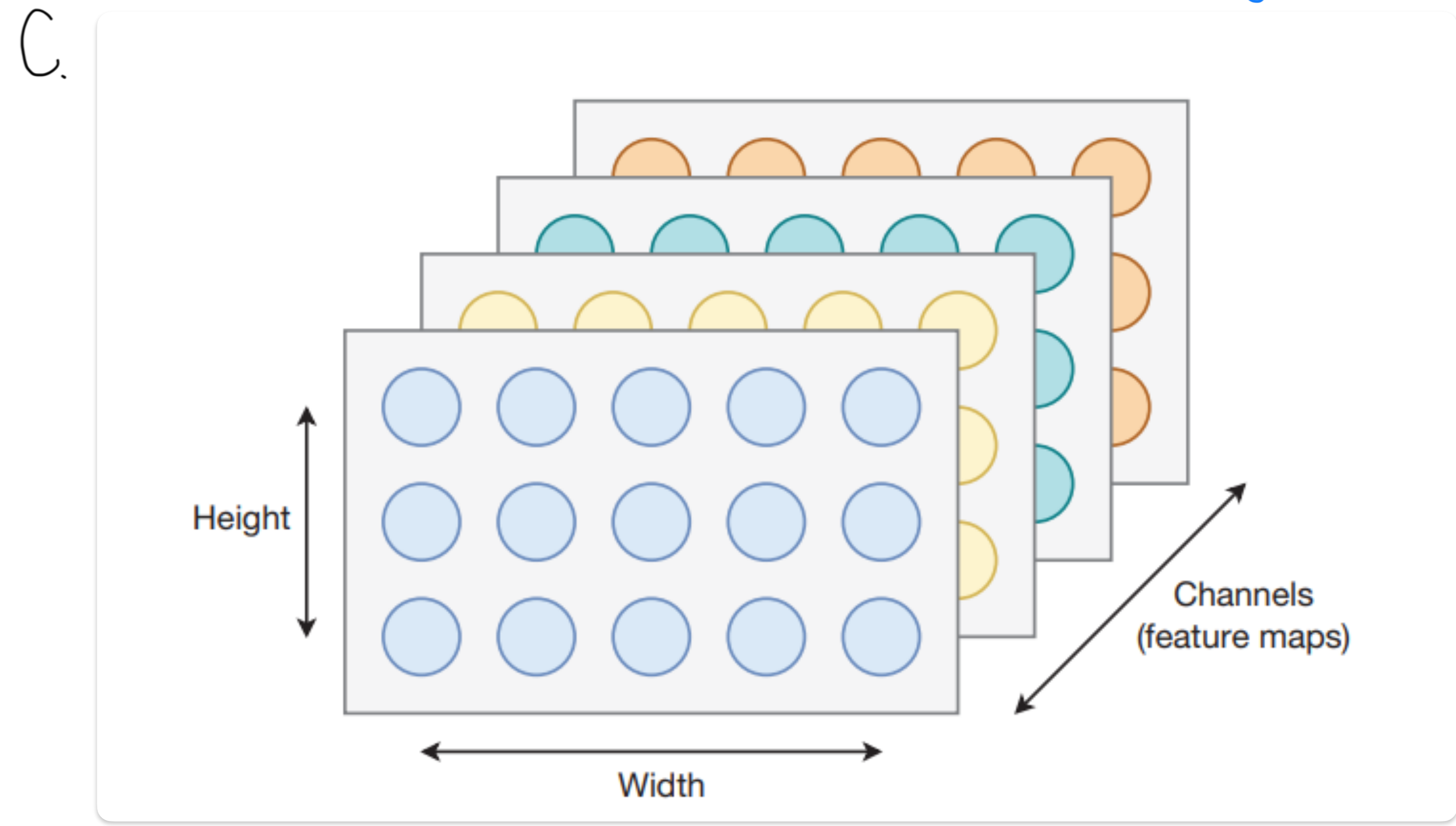
2. Affine Transformation:

- A. translation is a type of affine transformation, it changes the location of an object without changing its shape.
the blue rectangle represents a translated version of the red rectangle in Figure 7-3.
- B. rotation is another type of affine transformation, which change the orientation of an object.
the green rectangle represents a translated version of the red rectangle in Figure 7-3.



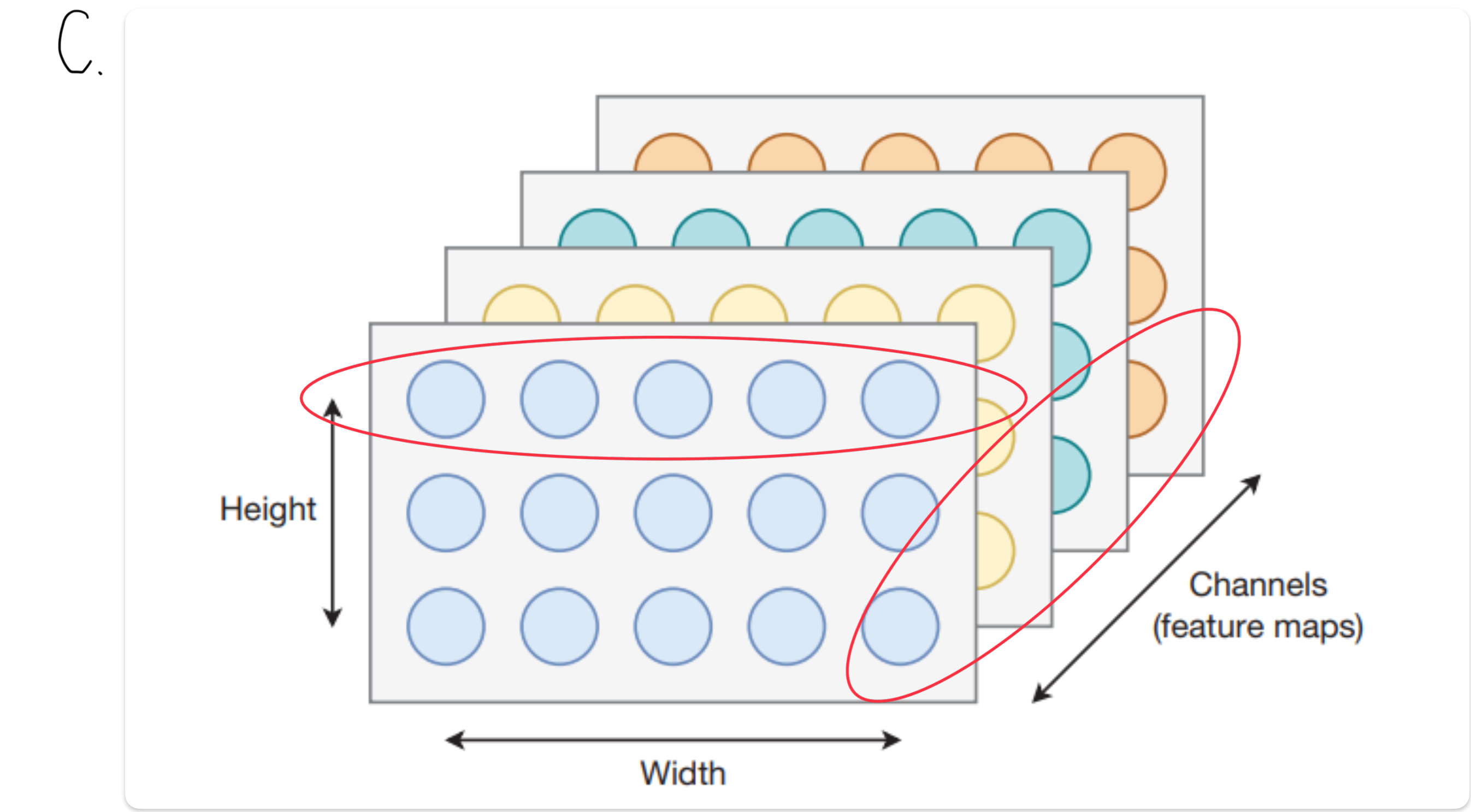
3. A Feature Map:

- A. a feature map is a 2D structure ($\text{Width} \times \text{Height}$), the width and height represent the grid dimensions of the neurons arranged within the feature map.
a single feature map represents the identification of one specific feature.
- B. all the neurons within the same feature map have an identical set of weights due to weight sharing.
that is, all the neurons within the same color in Figure 7-4 (= within the same feature map) are identical copies of each other, but they will receive different input values.



4. A Convolutional Layer:

- A. a convolutional layer is a 3D structure ($\text{Width} \times \text{Height} \times \text{Count}$), the width and height represent the grid dimensions of the neurons arranged within the feature map while the Count equals the num of output channels in a convolutional layer.
therefore, a convolutional layer is essentially composed of multiple stacked feature maps.
- B. all neurons within a convolutional layer are decoupled, meaning there are no connections among each neuron in a convolutional layer, like the state in Figure 7-4



5. A Convolution Kernel (=Kernel=Filter):

- A. we can combine the input weights of a single neuron in a convolutional layer into a 3D matrix.
we call this 3D matrix a convolutional kernel, which functions as a filter.
- B. we use a convolutional kernel as our smallest building block when describing a convolutional layer.
- C. due to the weight sharing mechanism within the same feature map, we obtain only one type of convolutional kernel for each feature map.

6. A Receptive Field:

- the region of pixels from a subset of a image that inputs into a single neuron is know as the receptive field of that neuron.